

TrES-3b

R-2815

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120609 · created 2026-09-05T01:52:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456087.7173 +/- 0.00097 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.02
Transit depth (Rp/Rs)^2	3.86 +/- 0.8 [%]
Orbital Inclination (inc)	81.5 +/- 0.54
Airmass coefficient 1 (a1)	1.019 +/- 0.011
Airmass coefficient 2 (a2)	-0.0148 +/- 0.008
Scatter in the residuals of the lightcurve fit is	1.09 %
Best Comparison Star	#3 - [540, 391]
Optimal Method	PSF photometry
Transit Duration (day)	0.0554 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.196 ± 0.02 vs 0.16309 (deviation 1.65σ)
Duration fit vs published	0.0554 d vs 0.0567 d (deviation 0.21σ)
Mid-time O-C	-2.2 min
Depth SNR	9.8
Residual scatter	1.09 %

Light curve

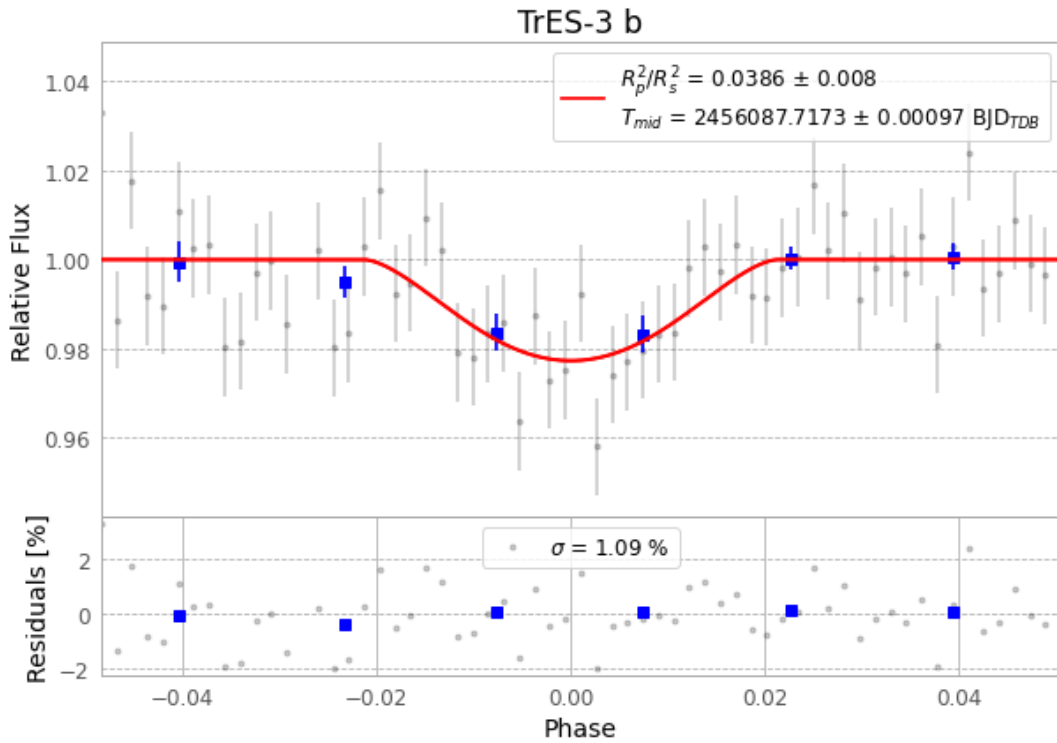


Figure 1 — FinalLightCurve_TrES-3 b_2012-06-08.png (SHA-256 155ebc74f172ad02...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cdc7a0ec3ab289ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), TRES-3120609033612.FITS → TRES-3120609064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120609.log (3682ed4e4f45...), FinalLightCurve_TrES-3 b_2012-06-08.png (155ebc74f172...), FinalLightCurve_TrES-3 b_2012-06-08.csv (93bf91dba914...)

Compute resources

Wall time 410.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 01:52Z.